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SUPERSEDING
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MILITARY SPECIFICATION

# CARRIER TELEGRAPH TERMINAL EQUIPMENT AN/UCC-1C(V)

## 1. SCOPE

1.1 This specification describes multichannel voice-frequency carrier telegraph terminal equipment, intended for use on long-haul radio circuits, wire lines and line-of-sight link circuits. The equipment shall make use of microelectronic integrated circuit devices and other state-of-the-art electronic techniques to achieve maximum performance, reliability and maintainability at minimum equipment cost. The equipment shall utilize modular techniques, so that, by the combination of basic modules, a terminal of desired channel capacity, input and output characteristics, and mode of operation may be obtained. The equipment shall include enclosures which permit self-contained mounting of the equipment by stacking of enclosures, which also contain provisions for relay rack mounting, thus providing for both shipboard and shore station installation. While reduction of equipment size through the use of microelectronic techniques is a desirable objective, size reduction is subordinate to attainment of maximum performance and reliability at lowest cost, considering such specific factors as repairability versus replacement of modules, trouble recognition and fault isolation, connector design, necessary panel space for controls. The equipment shall be physically and functionally interchangeable, at the system level, with earlier transistorized versions of the AN/UCC-1. The equipment shall be compatible with the AN/FGC-29, AN/FCC-16, AN/FGC-61, AN/FCC-3, and AN/FGC-60(V) Carrier Telegraph Terminal equipments.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

| | | |
|---|---|---|
| MIL-S-901 | - | Shock Tests, H.I. (High Impact); Shipboard |
| MIL-E-16400 | - | Electronic Equipment, Naval Ship and Shore, General Specification |
| MIL-E-17555 | - | Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of. |
| MIL-F-18327 | - | Filters; High Pass, Low Pass, Band Pass, Band Suppression, and Dual Functioning, General Specification for. |
| MIL-I-45208 | - | Inspection System Requirements |

STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-108 | - | Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment. |
| MIL-STD-188 | - | Military Communication System Technical Standards. |
| MIL-STD-189 | - | Racks, Electrical Equipment, 19-Inch and Associated Panels. |



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| MIL-STD-415 | - | Test Points and Test Facilities for Electronic Systems and Associated Equipment Design, Standard for. |
| MIL-STD-470 | - | Maintainability Program Requirements for System Requirements |
| MIL-STD-471 | - | Maintainability Demonstration |
| MIL-STD-721 | - | Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors, and Safety. |
| MIL-STD-781 | - | Reliability Tests: Exponential Distribution. |
| MIL-STD-785 | - | Reliability Program for Systems and Equipment Development and Production. |
| MIL-STD-1345(EC) | - | Data, Measurement, in support of Maintenance, Calibration, and Repair of Electronic Equipment. |

HANDBOOKS

MILITARY

| | | |
|---|---|---|
| MIL-HDBK-217 | - | Reliability Stress and Failure Rate Data for Electronic Equipment. |
| MIL-HDBK-472 | - | Maintainability Prediction |

PUBLICATIONS

MILITARY

| | | |
|---|---|---|
| NAVSHIPS-0969-125-0250 | - | Preferred General Purpose Electronic Test Equipment. |
| NAVSHIPS 94324 | - | Maintainability Design Criteria Handbook |
| NAVSHIPS 94501 | - | Reliability Design Criteria Handbook |
| NAVPERS 15105 | - | Naval Enlisted Classification Manual |
| NAVPERS 18068 | - | Qualification for Advancement and Rating Manual |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.2).

3.2 Material and parts. Except as otherwise stated herein, and to the extent that a Government-furnished model and drawings are provided, subassemblies, modules, enclosures, removable printed circuit cards, shall be mechanically, electrically, and physically interchangeable with corresponding parts of the Government-furnished model.

3.2.1 Signal line inputs and outputs shall be solid-state devices. No mechanical-magnetic relays shall be employed.

3.2.2 Printed circuitry shall be used as extensively as possible.

3.3 General requirements. The equipment shall meet the requirements of MIL-E-16400 except as hereinafter specified.

3.3.1 Power source. The equipment shall operate within the requirements specified herein when connected to a source of 50 and 60 hertz (Hz) ±5 percent, single phase ac power at 115 or 230 volts ±10 percent.



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3.3.1.1 Fuses. Fuses with blown-fuse indicators visible on the front panel shall be provided in each side of the primary power circuit as well as one side of the primary power circuit of each individual channel. Blown-fuse indicators for individual channels shall be located on each individual channel module. Blown-fuse indicator for individual primary power shall be located on power module.

3.3.2 Enclosures. To provide for shipboard and shore installations, there shall be a standard enclosure supplied as required. The standard enclosure shall accommodate eight transmit or receive modules and one control-attenuator module. All enclosures shall be in accordance with MIL-STD-108 (Protected class) and shall be equipped with proper brackets which will be capable of withstanding the weight of the channel enclosure if it is desired to install in a 19-inch rack. Construction of each enclosure shall be such that any channel can be withdrawn for servicing or adjustment during operation without interruption to other operating channels. Each enclosure shall incorporate a power module with blown-fuse indicators as described in paragraph 3.3.1.1 located on the front panel. Hardware for properly attaching and bonding more than one of the enclosures together shall be supplied, as well as necessary interconnecting cables. Holes for attaching enclosures shall be drilled and threaded inserts shall be supplied on the bottom holes and matching holes on the top of the enclosure.

3.3.2.1 Eight-channel enclosure. This enclosure shall house eight channel modules, either transmit or receive, or both. The enclosure shall be constructed in such a manner as to permit vertical stacking of up to four such enclosures, in a manner satisfactory for shipboard installation. The enclosure shall be suitable for mounting in a standard 19-inch relay rack in accordance with MIL-STD-189 for shore installation. Each enclosure shall be provided with the proper L brackets which will allow rack mounting. The L brackets shall be removable by removal of machine screws. The enclosure shall provide the following functions:

(a) A holder for eight plug-in channel modules, either transmit, receive, or both, including all connections between the channel modules and external signal and power sources, and a holder for the attenuator module. The attenuator module shall house the master power on-off switch, "power on" indicator, fuses, constant-level amplifier, transmit group attenuator, and required line transformers (see 3.5.1.4 and 3.5.2.2).

(b) Programming for various systems configurations such as nondiversity, frequency diversity, space diversity, four-channel diversity, signal sense, individual or parallel input-output shall be provided by means of the enclosure connector wiring in conjunction with switches in each channel module. The design of the equipment shall be such that a screwdriver is the only tool required for installation, programming, and alignment. Alternating current interconnection between enclosures shall be provided.

(c) The enclosure shall house a test fixture which shall allow operation of individual modules outside and independent of the enclosure. Each enclosure shall incorporate a constant level amplifier which shall allow sixteen tones to be automatically adjusted when any one or any combination of tones is turned off. Design details of the constant level amplifier are subject to design analysis by the contractor and approval by the command or agency concerned.

(d) Each 8-channel enclosure shall contain a provision to allow switching between 2- and 4-channel diversity operation without requiring opening of the signal line or removing modules. Location and wiring details of the switches are subject to design analysis by the contractor and approval by the command or agency concerned.

(e) The 8-channel enclosure shall include the following:
   (1) Module extraction handles satisfactory to the command or agency concerned.
   (2) A.C. power interconnect between 8-channel enclosures.
   (3) Quick-disconnect connectors (standard parts, MS 3112 series) on the rear of the 8-channel enclosure.
   (4) Channel center-frequency markers, visible from the front of each transmit and receive module.



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(5) Interconnection circuits to facilitate diversity alignment.
(6) Individual on-off switch on each keyer front panel to remove the tone output from the composite tone signal.

3.3.2.2 Multiplex-demultiplex enclosure. An enclosure shall be provided for a multiplexer-demultiplexer specified in 3.4.3. This enclosure shall be constructed in the same manner as specified in 3.3.2.1 to permit vertical stacking, except that this enclosure shall always be installed at the top of the vertical stack of enclosures specified in 3.3.2.1.

3.3.3 Accessibility. The equipment shall be so designed that all maintenance and operating procedures may be performed from the front of the cabinet.

3.3.4 Temperature. The equipment shall meet the temperature requirements of class 4 of MIL-E-16400.

3.3.5 Tropicalization. Tropicalization will not be required.

3.3.6 Ventilation. A ventilating system shall be included in the cabinet only if required to meet the temperature requirements.

3.3.7 Test facilities. Suitable test points shall be provided to enable ease of operation and maintenance using MIL-STD-415 as a guide. Test facilities and specialized test equipment shall be included as part of a terminal and shall be contained in a separate enclosure.

3.3.7.1 Test data requirements. Test point measurement parameters and support equipment data shall be provided to the command or agency concerned in accordance with MIL-STD-1345(EC). (See 6.1.2)

3.3.7.2 Selection of GPTE. General purpose test equipment (GPTE) shall be selected from NAVSHIPS 0969-125-0250 - Preferred general purpose electronic test equipment. If a requirement exists for a GPTE item which is not listed in this NAVSHIPS publication, a notification shall be forwarded to NAVELEX (Code 0463) requesting permission to substitute the required test equipment giving the required measurement parameters along with any other justification.

3.3.8 Test points. The following test points shall be provided on the front panel of the equipment, in the form of a single multiple-contact connector on the front panel of each module, mating with a test cable connector privided as part of the terminal test equipment. The female connector half will be mounted on the module front panel, and will be selected to enable 0.080 diameter test probes to be inserted directly into individual contacts without damage to the test connector.

(a) Transmit (Keyer) unit
(1) D-C loop current monitor (two test points).
(2) Output tone level (two test points).
(3) Bias.
(4) Plus 12 volts.
(5) Plus 3 volts.
(6) Minus 6.8 volts.
(7) Signal common.
(b) Receive (Converter) unit
(1) D-C loop current monitor (two test points).
(2) AGC level.
(3) Discriminator A.
(4) Discriminator B.
(5) Discriminator drive level (balance).
(6) Tone input.
(7) Plus 12 volts.
(8) Signal common.



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(c) Control attenuator
  (1) Composite tone level out (two test points).
  (2) Plus 12 volts.
  (3) RCV.A.
  (4) RCV.B.
  (5) Signal common.
(d) Multiplexer-demultiplexer
  (1) Direct path (two test points).
  (2) Composite signal (two test points).
  (3) Translated path (two test points).

3.3.9 Reliability.

3.3.9.1 Quanitative reliability requirements. The carrier telegraph nondiversity terminal shall have a specified mean-time-between failure (MTBF), $\theta_o$, as defined in MIL-STD-781, of 10,000 hours per channel. The MTBF for any piece part, module or subassembly whose failure will result in the failure of two or more telegraph channels shall be given by the formula MTBF $\geq$ 10,000 hours xCF where CF is the number of channels that will fail if the piece part, module, or subassembly fails. A channel of a non-diversity terminal consists of the following contractor supplied modules:

1 - Converter, Frequency Shift, CV-1920A/UCC-1C(v)
1 - Control - Attenuator, C-6554A/UCC-1C(v)
1 - Keyer, Frequency Shift KY-558A/(P)/UCC-1C(v)

The Test Set, Telegraph, TS-2232 AN/UCC-1C(v) for alignment and test of other telegraph terminal equipment is not considered as part of a channel.

3.3.9.2 Reliability program plan. The supplier shall submit for approval by the command or agency concerned, a reliability program plan in accordance with and in sequence with the paragraphs of MIL-STD-785. All paragraphs of MIL-STD-785 shall be addressed and the applicability shall be noted. A preliminary reliability program plan shall be submitted as part of the technical proposal. (See 6.1.2)

3.3.9.3 Reliability apportionment. The supplier shall perform an apportionment of the Carrier Telegraph Non-Diversity Terminal Channel reliability requirements to the module level ie: 1-Keyer, Frequency Shift; 1-Converter, Frequency Shift, and 1-Control attenuator. The apportionment shall be submitted to the command or agency concerned for approval. (See 6.1.2)

3.3.9.4 Reliability prediction. The supplier shall perform and submit for approval by the command or agency concerned the following reliability prediction:

(a) Design prediction based on stress factors and part population in accordance with the requirements of MIL-HDBK-217. (See 6.1.2)

3.3.9.5 Source of failure rates. The primary source of failure rate data shall be MIL-HDBK-217. For items not included in MIL-HDBK-217, other sources of data may be used with the approval of the command or agency concerned. When such alternate sources of data are proposed, sufficient information shall be submitted to enable the command or agency concerned to evaluate the validity and applicability of the results derived therefrom.

3.3.9.6 Reliability design guides. The supplier shall make maximum use of the design guides contained in NAVSHIPS 94501, Reliability Design Handbook.

3.3.9.7 Definitions. Reliability and maintainability terms used herein are as defined in MIL-STD-721.

3.3.9.8 Reliability reports. Reports in accordance with MIL-STD-785 that provide a complete accounting of progress on each element defined by the reliability program plan, results achieved and status of actions to resolve problems and correct deficiencies shall be submitted to the command or agency concerned . (See 6.1.2)



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3.3.9.9 Reliability demonstration tests. The reliability testing shall be in accordance with paragraph 4.4.3 of this specification.

3.3.10 Maintainability

3.3.10.1 Quantitative maintainability requirement. The Carrier Telegraph Terminal shall have an equipment repair time (ERT) of no greater than 45 minutes. This includes identification of the module in which a malfunction occurs and the repair of the malfunction.

3.3.10.2 Maintainability program plan. The supplier shall prepare and submit for approval by the command or agency concerned a maintainability program plan in accordance with the requirements of MIL-STD-470. The maintainability program plan shall describe the tasks to be performed and the procedures for conducting and controlling maintainability program. The plan shall address each paragraph of MIL-STD-470 noting those paragraphs which are not applicable. A preliminary maintainability program plan shall be submitted as part of the technical proposal. (See 6.1.2)

3.3.10.3 Maintainability prediction. A maintainability prediction shall be performed by the supplier using Procedure II of MIL-HDBK-472. The supplier shall document and justify all assumptions and the applicability of all data used in the prediction. The prediction shall include mechanical, electromechanical, and electronic parts. The results of the prediction shall be compared to the required ERT and where the predicted values do not achieve the requirements the contractor shall accomplish such changes in design as are necessary to improve the predicted values sufficiently to achieve the requirements. The prediction shall be submitted to the command or agency concerned for approval. (See 6.1.2)

3.3.10.4 Maintainability design guides. The supplier shall make maximum use of the design guides contained in NAVSHIPS 94324, Maintainability Design Criteria Handbook.

3.3.10.5 Maintainability reports. Reports in accordance with MIL-STD-470 that provide a complete accounting of progress on each element of the plan, results achieved and status of actions to resolve problems and correct deficiencies shall be submitted to the command or agency concerned. (See 6.1.2)

3.3.10.6 Maintainability demonstration tests. The maintainability requirements of the Carrier Telegraph Terminal shall be demonstrated in accordance with paragraph 4.4.4 of this specification.

3.3.10.7 Maintenance philosophy

3.3.10.7.1 The equipment maintenance concept is for maximum maintenance at the organizational level. The equipments are to be modular designed and constructed to permit maintenance in stages as defined below:

(a) Primary corrective maintenance: The maintenance necessary for removal and replacement of plug-in type assemblies or printed circuit cards for immediate fault correction of equipment and return to operational readiness.

(b) Secondary corrective maintenance: That maintenance and repair necessary to return the plug-in type assemblies or printed circuit cards removed from the equipment in the primary corrective maintenance stage to operational readiness. Secondary corrective maintenance will be maximized by limiting the complexity of the plug-in assemblies and printed cards to a degree such that the qualifications and capabilities of a NAVY technical enlisted specialist (E4/E5) will be able to effect the maintenance. When the complexity of the assemblies, modules, and cards is beyond these capabilities, it is incumbent upon the contractor to submit to NAVELEX, recommendations for the skill, capability, and maintenance level required.

3.3.10.7.2 Organizational maintenance. This level of maintenance is defined as the maintenance which is the responsibility of and performed by organizational personnel with assigned equipments. It normally consists of inspecting, servicing, lubricating, adjusting, and replacement of parts, printed circuit cards,



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subassemblies, and assemblies. Complexity of effort is not beyond the capability of a Navy technical enlisted specialist (E3/E4).

3.3.10.7.3 Maintenance personnel. Qualifications and capabilities for Navy enlisted personnel (see 3.3.10.7.2) shall be in accordance with NAVPERS 18068 and NAVPERS 15105 documents.

3.4 Equipment composition. The equipment shall be constructed in modular form to permit the derivation of a variety of required terminal configurations by use of building block technique. In addition to the enclosures for shipboard and shore installation specified in 3.3.2, the modules specified in 3.4.1 through 3.4.4 shall be provided as required. Transmit channel modules, receive channel modules and multiplex-demultiplex modules shall be provided to permit operation on narrow band channels 1 through 16 and wide-band channels 17 through 20. Midband frequencies shall be as shown in Table I. When frequency-diversity mode of operation is used, the channel combinations shown in Tables II and III shall apply:

Table I. Midband channel frequencies.

| Channel | Frequency nominal (Hz) |
|---|---|
| 2 | 425 |
| 4 | 595 |
| 6 | 765 |
| 8 | 935 |
| 10 | 1105 |
| 12 | 1275 |
| 14 | 1445 |
| 16 | 1615 |
| 1 | 1785 |
| 3 | 1955 |
| 5 | 2125 |
| 7 | 2295 |
| 9 | 2465 |
| 11 | 2635 |
| 13 | 2805 |
| 15 | 2975 |
| 17 1/ | 1955 |
| 18 1/ | 2380 |
| 19 1/ | 2805 |
| 20 1/ | 3230 |

1/ Wide band channels.

Table II. Possible narrow band diversity combinations.

| Channel | Frequency nominal (Hz) |
|---|---|
| 2 and 1 | 425 and 1785 |
| 4 and 3 | 595 and 1955 |
| 6 and 5 | 765 and 2125 |
| 8 and 7 | 935 and 2295 |
| 10 and 9 | 1105 and 2465 |
| 12 and 11 | 1275 and 2635 |
| 14 and 13 | 1445 and 2805 |
| 16 and 15 | 1615 and 2975 |



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Table III. Possible narrow and wide band diversity combinations.

| Channel | Frequency nominal (Hz) |
|---|---|
| 2 and 10 | 425 and 1105 |
| 4 and 14 | 595 and 1445 |
| 6 and 12 | 765 and 1275 |
| 8 and 16 | 935 and 1615 |
| 17 and 19 | 1955 and 2805 |
| 18 and 20 | 2380 and 3230 |

3.4.1 Transmit channel (Keyer) module. This module shall consist of a complete self-contained single channel FSK tone telegraph transmit unit including power supply. Tone generation circuits shall be of the crystal oscillator count-down type, to provide high frequency accuracy. All frequency-determining elements shall be contained in sealed assemblies which are securely attached to the module structure to withstand shock and vibration. All active circuit components shall be contained on plug-in printed circuit boards which are completely interchangeable between all transmit modules, narrow or wide band. The module shall include its own power supply. Designation of channel center frequency shall be permanently affixed to the front panel of each channel module and shall be visible from the front of the unit. Space shall be provided for temporary marking of channel numbers using grease pencil or other similar marking. Channel programming switch function indications shall be permanently provided by engraving or other method specified by MIL-E-16400.

3.4.2 Receive channel (Converter) module. This module shall consist of a complete single channel FSK tone telegraph receive unit capable of nondiversity, space or frequency diversity modules of operation, or both. The construction techniques shall be similar to the transmit channel module. Receive channel modules shall be physically interchangeable with the transmit channel modules. Detection circuits shall utilize conventional frequency discriminator techniques for reliability and proven performance. Frequency determining components shall be attached to the module structure in a manner which permits rapid removal, yet which provides the necessary rigidity to withstand shock and vibration.

3.4.3 Multiplexer-demultiplexer module. This module shall consist of a complete multiplexer-demultiplexer which shall provide means of transmitting or receiving telegraph, telephone and facsimile signals simultaneously over independent sideband radio circuits. It shall make use of microelectronic integrated circuit devices to the maximum extent consistent with improvements in reliability and maintainability over preceding versions. The multiplexer-demultiplexer module shall be capable of accepting two nominal 300 to 3000 Hz voice frequency channels and converting them into a single 300 to 6000 Hz channel suitable for radio transmission over a 6 kHz independent sideband transmitter. As a demultiplexer, it must accept a 300 to 6000 kHz audio channel from a radio receiver and convert this information into its original 3 kHz channels.

3.4.3.1 All active circuitry shall be mounted on plug-in repairable printed circuit cards replaceable as an assembly.

3.4.3.2 A fixture shall be provided for operating and testing individual transmit or receive channel modules outside and independent of the enclosure. This fixture shall be provided for each enclosure and shall be stored within and be conveniently detached from the enclosure.

3.4.4 Test equipment module. This module shall consist of a completely self-contained unit including power supply and shall be housed in a portable enclosure in accordance with MIL-STD-108 (Protected class). The unit shall provide the following test facilities:

(a) Tone level measurement of zero dbm, to an accuracy of ±1 db.
(b) Loop current measurement to an accuracy of ±5 ma.
(c) Reversal generator for d-c telegraph signals, including mark-space at 75 and 150 baud. Both neutral and low level polar reversals shall be provided.



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(d) Receive phase distortion measurement to an accuracy of ±3 percent.
(e) Audible signal tracer (speaker and phone jacks shall be provided).
(f) A manually switchable tone generator which shall generate each of the 20 FSK tones in Table I. The levels, baud rates, and frequencies shall be those required of the AN/UCC-1 keyers, as specified herein, and shall be suitable for checking out a receive only configuration of the AN/UCC-1. The tone stability of the test set tone generator shall be within plus or minus 3 Hz. Automatically-keyed mark-space reversals are not required. The mark and space tones of each channel shall be manually selected by a switch or switches as necessary.
(g) A three-position power switch shall be provided to select and control the a-c power voltage desired (115 or 230 volts). The power switch shall be provided with a guard to permit only the "OFF" position and one input voltage to be selectable when the guard is in place. Removal and repositioning of the guard shall be necessary in order to change from 115 to 230 volts, and vice versa.
(h) Connection to the module under test will be made by a test cable, attached to the test equipment module and terminating in a male connector which mates with the test connector located on the front panel of each keyer, converter, and control-attenuator module.

3.4.4.1 The test equipment module shall have the maximum dimensions of 6 by 8 by 7 inches. It shall comply with the shock and vibration requirements of the basic equipment. The test equipment enclosure shall be capable of being bolted or attached to any of the basic enclosures specified herein. It shall also contain a handle for portable use. Any increase in the above limiting dimensions required due to added functions or for any other reason, is subject to approval by the command or agency concerned.

3.5 Equipment characteristics.

3.5.1 Transmit group.

3.5.1.1 Keying rate. The transmit modules shall be capable of handling keying at a minimum modulation rate of 75 baud for nonsynchronous operation on the narrow band channels 1 through 16 and 150 baud for synchronous operation on the wide band channels 17 through 20.

3.5.1.2 Frequency modulation. The frequency deviation from midband for the conditions of keying (mark and space) shall be +42.5 Hz and -42.5 Hz for the narrow-band channels and -85 Hz and -85 Hz for the wide-band channels. (Note: A positive sense indicates that the transmitted tone on a marking telegraph input signal is above the center frequency, whereas a negative sense indicates that the marking is below the center frequency.)

3.5.1.3 Input circuits. The transmit modules shall be capable of being keyed from the following sources:

(a) Standard interface (Low Level) in accordance with the applicable paragraph of MIL-STD-188 (battery supplies from the external loop).
(b) 20 to 60 milliamperes (ma) neutral telegraph loop battery 120 volts d-c supplied from the external loop.

3.5.1.3.1 The input circuits shall have maximum resistance of 100 ohms when used with a 60-ma loop and a maximum resistance of 300 ohms when used with a 20-ma loop.

3.5.1.3.2 Input circuit protection shall be provided in each transmit unit to protect against 250 volts 120 ma maximum overload.



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3.5.1.4 Output circuits. The output circuits of each of the transmit modules shall be balanced at the channel center frequency with respect to the ungrounded center tap of an output transformer, and shall be suitable for operation, either individually or in parallel with one or more of the other output circuits into a telephone line, radio transmitter or other device having a nominal 600-ohm impedance.

3.5.1.4.1 The output level of each channel shall be continuously adjustable over the range from -24 to +6 dbm (referred to 1 milliwatt in 600 ohms). The harmonic content of any keyed tone shall be suppressed 50 db below the fundamental frequency.

3.5.1.5 The difference in level between mark and space signals in the same channel shall not exceed 2 db. Adjacent channel mark and space frequencies shall be attenuated 45 ±2 db. Attenuation at 170 Hz from center frequency shall be at least 50 db.

3.5.1.6 The frequency stability of the individual channels shall be such that mark and space frequencies are maintained within limits of ±0.5 Hz from their nominal values over a temperature range of 0 to 50 degrees C.

3.5.1.7 Control-attenuator power module. Each enclosure shall contain a control-attenuator power module which will provide the following functions:

(1) Transmitting group attenuator.
(2) Power "ON-OFF" switch, fuses and blown-fuse indicators for the enclosure.
(3) Constant-level amplifier.
(4) Fusing and test points for the constant-level amplifier.
(5) A test fixture for operating and testing individual modules independent from the enclosure.

The control attenuator shall provide all the necessary facilities for accepting the output signal from one or more of the transmit channels, where the total signal level at the input to the attenuator is between -24 dbm and +18 dbm inclusive and shall provide a common output circuit which shall be ungrounded and suitable for operation into a telephone line, radio transmitter or other device having a nominal 600-ohm input impedance.

3.5.1.7.1 Input circuit. An input circuit shall be provided with a nominal 600-ohm impedance capable of accepting the output signals of up to 16 narrow band transmit channels or 8 narrow band and 4 wide band transmit channels simultaneously.

3.5.1.7.2 Output circuit. The control attenuator located in each enclosure shall provide a common output circuit which shall be ungrounded and suitable for operation into a telephone line, radio transmitter or other device having a nominal 600-ohm input impedance.

3.5.1.7.3 Transmitting group attenuator. It shall be possible to adjust the total signal level at the output terminals of the group attenuator to any level between -38 dbm and +6 dbm inclusive in increments not greater than 2 db. Adjustment of signal level shall be possible without interruption of equipment operation.

3.5.1.7.4 Power control. A three-position a-c power switch and a fuse for each leg of the primary power source to the enclosure shall be provided. Fuses shall have a blown-fuse indicator mounted on the front panel. The power switch shall be provided with a guard which can be positioned to permit only the "OFF" position and one a-c input voltage to be used when the guard is in place. Removal and repositioning of the guard shall be necessary in order to change from 115 to 230 volts, and vice versa.

3.5.1.7.5 Constant-level amplifier. A constant-level amplifier will be provided within the control attenuator which will maintain the output level of the control attenuator within 2 db of any preset value with the requirements of -38 dbm and +6 db. The constant output level shall be automatically maintained within this range when any number of tones between one and 16 is applied to the input, within the limits of -24 dbm to +18 dbm.

3.5.1.7.6 Fusing and test points (constant-level amplifier). Test points available on the panel of the control attenuator shall be those specified in 3.3.8(c). A blown-fuse indicator for the constant-level amplifier shall be provided on the front panel of the control-attenuator module.

3.5.1.7.7 Test fixtures. A fixture shall be provided for operating and testing individual transmit, receive, and control-attenuator modules outside and independent of the enclosure. This fixture shall be provided for each enclosure and shall be stored within and be conveniently detached from the control attenuator.



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3.5.2 Receive group.

3.5.2.1 General requirements. The receive channel modules shall be designed to operate with the transmit modules specified in 3.5.1. The receive modules shall have corresponding midband channel frequencies, keying rate capabilities and frequency modulation characteristics and shall be physically interchangeable with the transmit channel modules.

3.5.2.2 Input circuits. The input circuits of each of the receive modules shall be balanced with respect to the ungrounded center tap of an input transformer, and shall be suitable for operation either individually or in parallel with one or more of the other input circuits, from a telephone line, demultiplexer, radio receiver, or other device having a nominal 600-ohm impedance.

3.5.2.3 Output circuits. Each receive module shall have a maximum output impedance of 800 ohms, capable of keying the following:

(a) 20 to 60 ma neutral telegraph loop, either side grounded with battery (120 volts d-c) supplied from the external loop.
(b) 30 to 130 volts d-c, 20,000 ohms minimum input impedance circuit positive or negative grounded with battery supplied from external loop.
(c) Standard interface (Low Level) in accordance with the applicable paragraph of MIL-STD-188 (battery supplied from the external loop).

3.5.2.3.1 Protection against circuit failure other than fuse blow out shall be provided in cases where currents in excess of 120 ma and voltages up to 300 volts are applied to the loop. Protection shall also be provided in cases where reverse polarity of up to 400 volts is applied to the loop. The output circuits shall be such that they cannot be damaged by any combination of signal level and control settings allowable under this specification.

3.5.2.4 Frequency and space diversity. A receive module shall be capable of operating in either non-diversity or in frequency or space diversity. The combining of two or four receive channels in diversity shall be accomplished by equal gain diversity techniques in such a manner that one d-c telegraph output results and the actuation of this output results from the contribution of the two (or four) receive signals in proportion to their relative strengths.

3.5.2.5 Input level. When the phase distortion control (see 3.5.2.6) is turned to its minimum delay position, any receive channel shall perform within the requirements of this specification without readjustment when receiving a signal from the corresponding transmit channel at any level in the range from -40 to +10 dbm or when the receive signal is varying slowly within this range. When the phase distortion control is turned to its maximum delay position, the lower operating signal limit may be decreased to -35 dbm. Input filter requirements shall be similar to those specified in 3.5.1.5 and 3.5.1.6. In addition attenuation at 170 Hz from center frequency shall be at least 50 dbm.

3.5.2.6 Phase distortion control. Each narrow-band recieve module shall include a continuously variable time delay circuit which will cause a delay range of the receive signal of at least 0.0035 second in not less than 270 degrees of rotation of the control, and without any additional adverse alteration at the signal. The required delay characteristics for each wide-band receive channel shall be the same as for the narrow-band channels except that the delay shall be from zero to at least 0.0017 second. The delay adjust circuit shall be such that, in each receive module, when the delay adjust control is turned fully in the direction for maximum delay, the delay shall be as follows:

(a) Narrowband channels - - - - - - - - - - - - - 0.0035 seconds plus or minus 5 percent.
(b) Wideband channels - - - - - - - - - - - - - 0.0017 seconds plus or minus 5 percent.

3.5.3 Multiplexer-demultiplexer module.

3.5.3.1 Input. The module when used as a multiplexer shall be capable of handling two separate voice frequency (vf) input circuits each having a bandwidth of 375 to 3025 Hz and an impedance of 600 ohms.



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The input shall be capable of accepting signals from carrier telegraph units having levels of a maximum of +4 dbm per channel and +13 dbm for a single frequency.

3.5.3.2 Output. The multiplexer shall derive a single complete output from the two input signals by means of frequency division multiplexing. The bandwidth of the output signal shall be from 325 to 5915 Hz. The output signal shall have a maximum single frequency level of +22 dbm and a composite carrier telegraph signal level of +5 dbm (32 channels) per channel.

3.5.3.3 Gain. The gain of the multiplexer shall be 24 db ±1 db over the entire frequency range, adjustable in steps of 1 db.

3.5.3.4 Input and output impedance. The input and output circuits shall be provided with a balanced nominal 600-ohm impedance.

3.5.3.5 The demultiplexer portion of the module shall be designed to operate with and as part of the multiplexer in the same enclosure. The module when used as a demultiplexer shall be capable of developing two output signals which shall be essentially identical to the two vf input circuits fed into the multiplexer.

3.5.3.6 Input. The demultiplexer shall be capable of handling a single vf input circuit having a bandwidth of 325 to 5915 Hz and an impedance of 600 ohms. The input shall be capable of accepting carrier telegraph signals having levels up to +13 dbm per channel and up to +5 dbm composite carrier telegraph (16 channels).

3.5.3.7 Output. The demultiplexer shall derive two outputs from the single input signals by means of frequency division multiplexing. The bandwidth of the output signals shall be from 325 to 3025 Hz. The output signal shall have a maximum level of -3 dbm per carrier telegraph channel and +12 dbm for a single frequency.

3.5.3.8 Gain. The gain of the demultiplexer shall be 24 db, adjustable in steps of 1 db.

3.5.3.9 Input and output impedance. The input and output circuits shall be provided with a nominal 600-ohms impedance.

3.5.4 Filters, receive. Receive bandpass filters of the AN/UCC-1C(V) shall meet all requirements of grade 4, class P, life expectancy X of MIL-F-18327, except as specifically noted herein.

3.5.4.1 Electrical requirements. Electrical requirements shall be as follows:

(a) The reference frequencies shall be as specified in Table I.
(b) The insertion loss measured at the reference frequency between a 600-ohm resistive unbalanced source and the internal load at an available source power of 0 dbm and with R equal to zero ohms shall not exceed 12 db, where R refers to the time delay adjustment resistance.
(c) The attenuation measured in accordance with the conditions of (b) herein at ±42.5 Hz from the reference frequency shall not exceed 2.5 db at +25 degrees C and shall differ by not more than 1 db.
(d) The attenuation measured in accordance with the conditions of (b) herein at ±85 Hz from the reference frequency shall be not less than 18 db.
(e) The attenuation, measured in accordance with the conditions of (b) herein at ±125 Hz and greater from the reference frequency, shall be not less than 45 db.
(f) The input admittance of the filter, measured at 125 Hz and further below the reference frequency and at 125 to 10,000 Hz above the reference frequency shall not exceed 333 ±J 1333 micromhos.
(g) When excited by an AM pulse train, the amplitude of the decaying envelope measured across the filters output with R equal to zero shall not exceed 4 percent of the "on" amplitude 12.5 to 27 milliseconds after the envelope has decayed to half amplitude. The source shall be 600 ohms



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unbalanced providing a rectangular keyed AM envelope having a carrier frequency equal to the reference frequency "on" "off" periods of at least 35 milliseconds and delivering "on" power of +6 dbm and -25 dbm.

(h) The 45-db bandwidth measured between a 600-ohm resistive unbalanced source and the internal load with an available source power of +6 dbm and -25 dbm shall be maintained within ±6 Hz of the 45-db bandwidth measured with an available power of 0 dbm. The mean frequency between the 45-db frequencies shall be maintained within ±3 Hz under the same conditions of measurement.

(i) The 45-db bandwidth measured between a 600-ohm resistive unbalanced source and the internal load with an available source power of 0 dbm at temperature of 0 degrees C and +60 degrees C shall be maintained within ±6 Hz of the 45-db bandwidth measured at +27 degrees C. The mean frequency between the 45-db frequencies shall be maintained within ±3 Hz under the same measurement conditions.

(j) The filter shall withstand application of a potential of 500 volts rms 60 hertz between terminals and case and between individual terminals. At an input power level of zero dbm and a temperature of +25 degrees C, the filter delay, defined as delta theta of delta f, and calculated from the measured frequencies for a ±90 degree phase shift, shall be 7.95 ±0.30 milliseconds when delay adjustment resistance (R) equals 100K ohms. For R equal to 50 ohms, the delay, calculated as above, shall increase by 4.0 milliseconds, minimum.

3.5.4.2 Mechanical requirements. Mechanical requirements shall be as follows:

(a) Dimensions. The receive filters shall have the following maximum dimensions:
  (1) Width - 1-7/16 inches
  (2) Depth - 6 inches
  (3) Height - 3-1/4 inches
(b) Mounting. Mounting shall be by means of four tapped inserts located on the base surface.
(c) Terminals. A 90GFR/40W-TP-8B Electrical Industries type header, or equivalent, shall be used and shall be located on the surface opposite the mounting surface.
(d) Weight. Weight of filters shall be minimized. Maximum weight shall be 35 ounces.

3.5.4.3 Marking. Marking of filters shall be in accordance with MIL-F-18327.

3.5.4.4 Environmental requirements. Filters shall withstand environmental testing for class 4 equipment of MIL-E-16400.

3.5.5 Filters, transmit. Transmit bandpass filters of the AN/UCC-1 shall meet all requirements of grade 4, class P, life expectancy X of MIL-F-18327, except as specifically noted herein.

3.5.5.1 Electrical requirements. Electrical requirements shall be as follows:

(a) The reference frequencies shall be as specified in Table I.
(b) The insertion loss, measured between 600-ohm resistive unbalanced source and load impedance at an output level of zero dbm shall not exceed 8 db at the channel center frequency.
(c) At 25 degrees C the attenuation measured in accordance with the conditions of (b) herein, at ±42.5 Hz from the reference frequency shall not exceed 2 db and shall differ by not more than 1.0 db.
(d) The attenuation, measured in accordance with the conditions of (b) herein at ±85 Hz from the reference frequency, shall be not less than 9 db.
(e) The attenuation measured in accordance with the conditions of (b) herein at ±125 Hz and greater from the reference frequency, shall be not less than 18 db.
(f) The attenuation measured in accordance with the conditions of (b) herein at harmonics of the reference frequency up to 10 kilohertz shall exceed 50 db.
(g) The input impedance of the filter measured at harmonics of the reference frequency with the output terminated in 600 ohm shall have a magnitude not exceeding 50 ohms, except that 90 ohms shall be allowed on the 425-hertz band.



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(h) The output admittance of the filter, measured at -125 Hz and further below reference frequency and at 125 to 10,000 Hz above reference frequency with the input terminated in 600 ohms shall not exceed 333 ±J 1333 micromhos.

(i) When excited by an AM pulse train, the amplitude of the decaying envelope measured across the filter load 12 to 27 milliseconds after the envelope has decayed to half amplitude shall not exceed 3-1/2 percent of the "on" amplitude. These limits shall be met with the filter delivering "on" power between +6 dbm and -25 dbm to a 600-ohm unbalanced load. The source shall be 600 ohms unbalanced providing a rectangular keyed AM envelope having a carrier frequency equal to the reference frequency and "on" and "off" periods of at least 35 milliseconds.

(j) The 20-db bandwidth measured between 600-ohm resistive unbalanced source and load impedances with an available source power of +16 dbm and -25 dbm shall be maintained within ±6 Hz of the 20 db bandwidth measured with an available source power of +10 dbm. The mean frequency between the 20-db frequencies shall be maintained within ±3 Hz under the same conditions of measurement.

(k) The 20-db bandwidth measured between 600-ohm resistive unbalanced source and load impedances with an available source power of 10 dbm at temperatures of 0 degrees C and +60 degrees C shall be maintained within ±6 Hz of the 20-db bandwidth measured at +27 degrees C. The mean frequency between the 20-db frequencies shall be maintained within ±3 Hz under the same measurement conditions.

(l) The filter shall withstand application of a potential of 500 volts rms, 60 hertz, between terminals and case.

(m) The delay defined as delta theta of delta f and calculated from the measured frequencies of ±90 degree phase shift shall be 7.6 ±0.3 milliseconds.

3.5.5.2 Mechanical requirements. Mechanical requirements shall be as follows:

(a) Dimensions. The transmit filters shall have the following maximum dimensions:
  (1) Width - 1-7/16 inches
  (2) Depth - 3-1/2 inches
  (3) Height - 3 inches minimum

(b) Mounting. Mounting shall be by means of four 6-32 tapped inserts located on the base surface.

(c) Terminals. Terminals shall be type 90GFR/40W-TP-6B, Electrical Industries, or equivalent, and shall be located on the surface opposite the mounting surface.

(d) Weight. Weight of filters shall be minimized. Maximum weight shall be 14 ounces.

3.5.5.3 Marking. Marking of filters shall be in accordance with MIL-F-18327.

3.5.5.4 Environmental requirements. Filters shall withstand environmental testing for class 4 equipment of MIL-E-16400.

3.6 System operating requirements.

3.6.1 Operating conditions. With a complete transmit terminal directly connected to a complete receive terminal to form a circuit from the send d-c telegraph circuits to the receive telegraph circuits, the equipment shall perform with the requirements specified herein under the operating conditions specified hereinafter.

3.6.1.1 Condition A. Condition A shall be the combined output circuits of the 16 transmit modules connected to the combined input circuits of the corresponding receive terminal through the untranslated paths of the multiplexer-demultiplexer units. Diversity combinations shown in Table II shall be used. The output level of each transmit channel shall be set to -10 dbm. The group attenuator shall be set at zero attenuation.

3.6.1.2 Condition B. Condition B shall be the same as condition A except that a 30-db loss shall be inserted between the transmit unit and the receive unit.



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3.6.1.3 Condition C. Condition C shall be the same as condition A except that the output circuits of the transmit terminal shall be directly connected to the combined input circuits of the receive terminal in accordance with the diversity combination shown in Table III.

3.6.1.4 Condition D. Condition D shall be the same as condition C except that a 30-db loss shall be inserted between the transmit unit and the receive unit.

3.6.1.5 Condition E. Condition E shall be the same as condition A except that the translated paths of the multiplexer and demultiplexer shall be used.

3.6.1.6 Condition F. Condition F shall be the same as condition B except that the translated paths of the multiplexer and demultiplexer shall be used.

3.6.2 Cross-talk limits. With the combined output circuits of 15 narrow-band transmit channels directly connected to the combined input circuits of a complete 16-channel receive terminal and with all transmit channels providing output levels of 0 dbm and being randomly keyed, it shall be possible to add the 16th transmit channel to the combined input circuit of the receive terminal through a 30-db, 600-ohm attenuator and maintain less than 40 percent total peak distortion as measured by the AN/GGM-1 or equivalent distortion test set or dual trace scope.

3.6.3 Telegraph signal distortion limits. With the converter filter delay control set to give minimum delay, with equipment operating under conditions A, B, C, D, E, or F, and with all channels carrying random signals keyed at their specified maximum rate, the over-all telegraph signal introduced by the equipment shall not exceed 5 percent total (peak) distortion.

3.6.4 Phase distortion limits. With the phase distortion controls specified in 3.5.2.6 set to give no delay, the phase distortion coefficient shall not exceed 0.002 second (2.0 milliseconds) for the frequency band specified herein and under operating condition A, B, C or D.

3.6.4.1 Phase distortion equalization. With the equipment operating under condition A, B, C, D, E, or F and with all transmitters keyed simultaneously, it shall be possible to equalize any signal delay, by means of the variable delay controls specified in 3.5.2.6 so that the phase distortion coefficient shall not exceed 0.0002 second(0.2 milliseconds) for the frequency band specified herein.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractors quality assurance. The supplier shall provide and maintain an effective inspection and quality assurance system acceptable to the Government. The inspection system shall conform to the requirements of MIL-I-45208. A current written description of the means by which the supplier will assure the quality of products shall be submitted to the cognizant Government inspector for evaluation. The written description shall include flow charts showing complete flow of material from inspection on receipt through all manufacturing processes to final shipment. The various process and inspection positions shall be clearly identified on the flow charts and cross-referenced to the detailed inspection procedures and criteria of the written description. Any changes to the procedures or the described system shall be submitted to the Government inspector for evaluation prior to use. (See 6.1.2)

4.1.2 Government verification. All quality assurance operations performed by the supplier will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations



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to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all requirements of this specification. Failure of the supplier to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Test procedures. The supplier shall submit written test procedures including test equipment and tolerance limits to be used for first article inspection and quality conformance inspection. Each test shall be identified in accordance with paragraph 4.2 as to classification. (See 6.1.2)

4.1.3.1 Procedures for inspections, tests, and demonstrations, shall contain step by step information which will enable the operator to properly perform the requirements of this specification. The method of examination shall be clearly defined.

4.1.3.2 The procedures shall contain a list of test equipment identified by name and model. Special test equipment shall also be listed and identified.

4.1.3.3 Data sheets shall be included in the procedures. Conformance parameters shall be clearly indicated with tolerance limits. Whenever quantitative measurements are made, the result shall be shown by quantitative data.

4.2 Classification of inspection. The method of examination and testing of the equipment fall within the following classifications:

(a) First article inspection
(b) Quality conformance inspection
    (1) Production inspection
    (2) Production control inspection
(c) Reliability testing
(d) Maintainability demonstration

4.3 First article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article tests shall be conducted as specified in MIL-E-16400 except that the following tests shall be omitted:

(a) Water cooling
(b) Accelerated Life Test
(c) Magnetic Test

4.3.1 The operating test specified in MIL-E-16400 shall be performed on each transmit, receive, and attenuator module; each test equipment module; each multiplexer-demultiplexer module; and on a system level. (See 4.3.3 for configuration of equipment for system level test).

4.3.2 Interchangeability. The supplier shall demonstrate physical and functional interchangeability of the modular units, the removable printed circuit cards, and the standard enclosures, between the supplier's equipments and with the government furnished equipment if furnished.

4.3.3 Equipment configuration for first article and quality conformance inspection. For first article and quality conformance tests, the equipment tested shall consist of four enclosures, four control-attenuators, sixteen keyers, and sixteen converters. The keyers and converters shall be operated in a back to back configuration.

4.3.4 First article inspection report. A first article inspection report shall be prepared containing all the results of the first article inspection. The report shall include completed Form NAVEXOS 3373 as specified in MIL-S-901 for the shock test and resilient mount data as specified in figure 3 of MIL-E-16400.



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The first article inspection report shall be submitted to the command or agency concerned, via the Government inspector, for review and approval. The first article inspection report shall indicate specification compliance limits as well as actual performance test results. The contractor shall certify that, on the basis of the report, the equipment does or does not comply with specification requirements. (See 6.1.2)

4.3.5 First article model acceptance. Acceptance of the first article model shall be contingent upon approval by the command or agency concerned of the first article inspection report.

4.3.5.1 Any further laboratory testing of the first article model may be made at the discretion of the Government. Additional testing shall not require more than 60 days.

4.4 Quality conformance inspection

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of adjustments, and tests which detect hidden defects of materials. Production inspection shall consist of the tests approved in accordance with paragraph 4.1.3 and shall include the requirements of Table I.

Table I. Production inspection

| Requirement | Reference |
|---|---|
| Surface examination | MIL-E-16400 |
| Operating test | MIL-E-16400 |

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to Table II below and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. Production control inspection shall be in accordance with MIL-E-16400 and shall be approved in accordance with 4.1.3. The equipment shall satisfactorily meet the requirements of Production Control Inspection prior to release for shipment.

Table II. Production Control Inspection

| Inspection | AQL% |
|---|---|
| Weights and Dimensions | 6.5 |
| Supply line voltage and frequency | 6.5 |
| Heat test | 6.5 |
| Power | 6.5 |

4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.2.2 Reinspection of conforming production control test units. Unless otherwise specified, sample units which have been subjected to and passed production control inspections may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all visible damage.

4.4.3 Reliability testing. The reliability of the carrier telegraph non-diversity terminal channel shall be verified and reported in accordance with MIL-STD-781. The specified MTBF, $\theta_0$, per channel, shall be 10,000 hours.



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4.4.3.1 Demonstration phase of production reliability tests. The demonstration phase of reliability tests shall be conducted in accordance with test plan IV at test level C. Voltage shall be cycled in accordance with the voltage cycling provisions of MIL-STD-781. The sample size shall be 2 (selected from the first months production) 16-channel non-diversity full duplex terminals consisting of 16 keyers, 16 converters, and 4 attenuators, each housed in 4 cabinets. The keyers and converters shall be operated in a back to back configuration. In the event that a procurement specifies terminals with less than 16-channels each then the number of terminals to be used for this test shall be increased to provide 32 channels as described above. The reliability requirements shall be successfully demonstrated prior to the shipment of any production equipment.

4.4.3.2 Production acceptance (sampling) phase of production reliability tests. The production acceptance (sampling) phase of reliability tests shall be conducted in accordance with test plan IV-A at test level A1, MIL-STD-781. The sample size shall be two terminals from each subsequent months production and shall be configured as described in paragraph 4.4.3.1. The reliability test shall be successfully completed prior to the shipment of each production lot.

4.4.3.3 Reliability test plan. The supplier shall prepare and submit to the command or agency concerned detailed reliability test plan in accordance with MIL-STD-785. Test parameters and conditions shall be in accordance with the equipment operational requirements as defined in the equipment specification. Performance parameters to be measured during the reliability test shall be in accordance with the equipment specification. Debugging and burn-in prior to the reliability test shall be limited to the debugging and burn-in to be performed on all production equipment prior to delivery to the command or agency concerned. No preventative maintenance shall be performed during a reliability test. Approval of the plan by the command or agency concerned must be received by the contractor prior to the start of reliability testing. (See 6.1.2)

4.4.3.4 Failure notification. The command or agency concerned shall be notified by telephone or TWX of failures occurring during the tests specified in paragraphs 4.4.3.1 and 4.4.3.2 that require repair or replacement of functional parts or assemblies. Telephone notification of failure shall be confirmed in writing. (See 6.1.2.)

4.4.3.5 Failure. Failure is defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth in this specification. All failures shall be considered relevant unless determined by the command or agency concerned to be caused by a condition external to the equipment under test which is not a test requirement and not encountered in service. Only relevant failures shall be counted during the reliability test for an accept or reject decision.

4.4.3.6 Final restoration and disposition of tested equipment. Upon completion of reliability testing, the tested equipment shall be returned to production for reinspection and reconditioning as required, and may be delivered as part of the contract quantity if it meets all terms and conditions of the contract for acceptance.

4.4.3.7 Reliability test reports. The supplier shall submit a report that defines the results of the test at the completion of reliability testing for approval by the command or agency concerned. (See 6.1.2)

4.4.4 Maintainability demonstration test. The maintainability requirements of paragraph 3.3.10.1 of this specification shall be successfully demonstrated with Method 3, MIL-STD-471. The sample size shall be as determined by the contractor and approved by the command or agency concerned. The maintainability requirements shall be successfully demonstrated prior to the shipment of any production equipment.

4.4.4.1 Maintainability demonstration test plan. The supplier shall prepare and submit for approval by command or agency concerned a maintainability demonstration test plan in accordance with MIL-STD-471. The plan must be approved by the command or agency concerned prior to the start of testing. (See 6.1.2)

4.4.4.2 Maintainability demonstration report. A final maintainability demonstration report shall be submitted in accordance with MIL-STD-471 at the completion of testing for approval by the command or agency concerned. (See 6.1.2)



1 Battery
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4.4.5 Inspection for preparation for delivery. Examinations shall be conducted to insure conformance with the requirements of Section 5 of this specification.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with MIL-E-17555 (See 6.1.1)

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

(a) Title, number and date of this specification.
(b) Number of first article samples to be submitted if other than specified in 4.3.3
(c) Marking, levels of preservation and packaging and packing required (see 5.1)

6.1.2 Contract data requirements. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423). The data required by this specification includes, but is not restricted to the following:

(a) Test data requirements (3.3.7.1)
(b) Reliability program plan (3.3.9.2)
(c) Reliability apportionment (3.3.9.3)
(d) Reliability prediction (3.3.9.4)
(e) Reliability reports (3.3.9.8)
(f) Maintainability program plan (3.3.10.2)
(g) Maintainability prediction (3.3.10.3)
(h) Maintainability reports (3.3.10.5)
(i) Contractors quality assurance (4.1.1)
(j) Test procedures (4.1.3)
(k) First article inspection report (4.3.4)
(l) Reliability test plan (4.4.3.3)
(m) Failure notification (4.4.3.4)
(n) Reliability test reports (4.4.3.7)
(o) Maintainability demonstration test plan (4.4.4.1)
(p) Maintainability demonstration report (4.4.4.2)

6.2 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
NAVY-EC

(Project 5805-N159)



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

| ORGANIZATION *(Of submitter)* | CITY AND STATE |
|---|---|

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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